Cartier–Miller evaluation of genus-one coefficient sums

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Revision 22 October 2026

Abstract

Let $p \geq 5$ be prime and let $f \in \mathbb{F}_p[x]$ be a squarefree cubic or quartic. We evaluate nonbranch reciprocal quadratic-character sums, and hence admissible base-field values of $f^{(p-1)/2} \bmod x^p$, by one differentiated Miller chain on a normalized quadratic twist. The underlying curve-order identity expresses the high polynomial block as a Laurent coefficient of a Miller logarithmic differential, without a coprimality or ordinarity hypothesis. The structural two-pole residue-divisor formula is retained for $p \geq 7$ and $f(0) = 1$, with a direct characteristic-p proof of its normalization at both exceptional traces. Supplied constant-plus-simple-pole weights and equivalent elliptic-point sums have explicit exclusion and infinity corrections. After exact-trace preparation, one query costs $O(\log p)$ field operations; field, bit and preparation costs are stated separately. The quarter-point binomial example and a rounded-third boundary illustrate the distinction between a fast prefix and additional arithmetic input. Correctness checks and historical matched-output benchmarks accompany the formulas.

1 Introduction

We study exact evaluation modulo a prime of coefficient sums attached to cubic and quartic polynomial powers. Although these powers have degree proportional to the prime, selected base-field values can be recovered through arithmetic on an elliptic curve. The cost of preparing the exact curve trace is separated throughout from the cost of each evaluation. Let $p \geq 7$ be prime, put $h = (p-1)/2$, and let $f \in \mathbb{F}_p[w]$ be squarefree of degree three or four, with $f(0) = 1$. Write

$$f(w)^h = \sum_m c_m w^m, \qquad T_f(w) = \sum_{m=0}^{p-1} c_m w^m, \qquad H = c_{p-1}. \tag{1}$$

For $\lambda \neq 0$ in $\mathbb{F}_p$, consider the weighted coefficient expression

$$S_f(\lambda) = \left[w^{p-1}\right] \frac{f(w)^h}{1-\lambda w} = \sum_{m=0}^{p-1} c_m \lambda^{p-1-m} = T_f\left(1/\lambda\right). \tag{2}$$

The last equality uses $\lambda^{p-1} = 1$ and is a base-field identity. At $\lambda = 0$, the coefficient expression gives $S_f(0) = H$; evaluation $T_f(0) = 1$ is separately immediate. Thus varying the rational weight amounts to evaluating one fixed truncation at different base-field arguments. We call an argument admissible when $\lambda = 0$, or when $\lambda \neq 0$ and $f(\lambda^{-1}) \neq 0$. The latter condition excludes the branch points of the double cover $v^2 = f(w)$. A polynomial with any nonzero constant term reduces to this normalization by dividing by that term and restoring its hth power afterwards. The pole normalization in Corollary 5 handles arbitrary constant terms directly. Euler's criterion also gives $T_f(\mu) = e - \sum_{m \geq p} c_m \mu^m$, where $\mu = \lambda^{-1}$ and $e = (f(\mu)/p)$ at a nonzero admissible argument.

The structural identity uses the following fixed geometric data. Let E be the smooth projective curve $v^2 = f(w)$ with origin $O = P_+ = (0, 1)$, invariant differential $\omega = dw/v$, and exact trace

$\tau = p + 1 - \#E(\mathbb{F}_p)$. For a nonzero admissible argument define

$$\begin{aligned} d = f(1/\lambda), \quad v_0^2 = d, \quad e = (d/p), \\ Q_\pm = (1/\lambda, \pm v_0), \quad \mathfrak{D} = (Q_+) - (Q_-), \quad r = \tfrac{1}{\lambda v_0}. \end{aligned} \tag{3}$$

For an integer M annihilating the class of $\mathfrak{D}$, fix a rational function $F_{\mathfrak{D}}$ with $\operatorname{div}(F_{\mathfrak{D}}) = M\mathfrak{D}$, and set $\Psi_M(\mathfrak{D}) = ((dF_{\mathfrak{D}}/F_{\mathfrak{D}})/\omega)(O)$. This value is independent of the scalar multiplying $F_{\mathfrak{D}}$. For $p \nmid M$ it retains the specified divisor representative; for $M = p$ it depends only on its divisor class.

Theorem 1 (Normalized residue divisor identity). Under these hypotheses, $H \equiv \tau \pmod{p}$, and every nonzero admissible argument satisfies

$$S_f(\lambda) = H - er\Psi_{M_e}(\mathfrak{D}), \qquad M_e = p + 1 - e\tau. \tag{4}$$

The product $r\Psi_{M_e}(\mathfrak{D})$ belongs to $\mathbb{F}_p$ and is unchanged by replacing v_0 by $-v_0$. The formula includes $\tau = e$, when $M_e = p$, and requires no division by that annihilator. The proof is given in Section 5.

The computational form is supplied by Theorem 2 and Theorem 4. If $W^h = U + H_E x^{p-1} + x^p J_W$ on a short model, the curve-order coefficient is $\alpha_{\#E(\mathbb{F}_p)}(a, b) = bJ_W(a)$, without a coprimality or ordinarity hypothesis. Normalizing at the requested nonbranch pole then gives one rational-point Miller chain on a twist. Corollary 5 recovers the original truncation from this chain. This is the recommended evaluator: it uses no quadratic algebra, splitting step or exceptional trace case distinction. Theorem 1 records the residue-divisor interpretation; its exceptional normalization remains a separate structural result.

The weighted-sum framework permits any squarefree cubic or quartic, including zero constant term and pole coordinate zero. The accelerated interface accepts a supplied constant-plus-simple-pole partial-fraction representation whose active poles are nonbranch. Theorem 6 accounts for all excluded fibres and gives the equivalent elliptic-point sums. Corollary 7 separates the $O(\log p)$ field-operation query from its bit cost and from trace acquisition. A trace-free selected-coefficient comparator uses classical polynomial-block machinery and also covers branch poles.

The binomial applications require a prefix and a boundary coefficient. At the quarter point classical CM data provide the boundary and trace. At the rounded third the trace is zero, while the boundary is equivalent to a modular factorial. The specified de Rham basis in Appendix B identifies that boundary without supplying a fast extractor. Detailed boundary backends, remainder forests, implementation policies and full benchmark protocols are collected in the accompanying computational report (Jordan, 2026e). The main paper retains the mathematical applications and a compact account of the measured workloads.

2 Relation to prior work

The geometric ingredients of the evaluation are classical. The Artin–Schreier construction underlying the high polynomial block is traced to Deuring (1941) in the accounts of Voloch (1990, Lemma 1.1; 1997, Section 2). Cartier (1957) established the differential operator used here, and Serre (1958, Propositions 8 and 10) describes its properties and the correspondence between geometric p-torsion divisor classes and Cartier-fixed holomorphic logarithmic differentials. These results provide the coefficient-selection and logarithmic-transfer framework for the present paper.

Cassels (1971, Sections 1 and 5–6) develops the high-block Artin–Schreier equation and its analogy with elliptic quasi-periods. Gunji (1976, equations (2.4), (2.9), and Lemma 2.11) studies functions with divisor $p((P) - (O))$, their logarithmic differentials and Cartier eigenvalues on ordinary elliptic curves. His setting is not restricted to trace-one curves. His differential is dx/y, whereas our short model uses $\omega_0 = dx/(2y)$; this factor must be retained when comparing scalars. The rounded-third de Rham interpretation also sits within the classical relationship between de Rham cohomology, Frobenius and Dieudonné theory developed by Oda (1969).

Trace acquisition is a separate classical algorithmic task. Schoof (1985) gives a deterministic polynomial-time point-counting algorithm; Schoof (1995) surveys the method and subsequent developments. This supplies the preparation guarantee used in Corollary 8. Our $O(\log p)$ field-operation bound concerns a query after an exact trace has been supplied or computed. It does not replace point counting or describe the bit cost of the complete calculation. The specialized CM preparation and the practical trace providers used in the experiments have their own costs and hypotheses.

Miller's foundational contribution is the short arithmetic representation of rational functions with prescribed divisors. His unpublished manuscript dated 6 May 1986 constructs these functions as straight-line programs through divisor addition, avoiding their full polynomial expansion (Miller, 1986, Section 2). The later published exposition (Miller, 2004, Section 4.1, especially equations 7–9) gives the divisor, addition and composition identities for the functions $f_{m,P}$, together with addition-chain evaluation in $O(\log m)$ elliptic additions. These are the functions and identities used in our differentiated recurrence and in Theorem 2. The 2004 publication is therefore cited as a later exposition of machinery already available before Semaev's 1998 and Rück's 1999 papers. Our computation extracts a local coefficient of $d \log f_{m,P}$ from that circuit; the short function construction itself is not a contribution of this paper.

Voloch (1990, Lemma 1.1 and Proposition 1.3) identifies the Artin–Schreier cover with Verschiebung and obtains an explicit p-descent homomorphism. Voloch (1997, Propositions 1–3) develops the characteristic-p zeta function, its addition law and its relation to logarithmic differentials, while correcting scale and sign conventions in the earlier account. On a trace-one short curve, the descent function is represented by the product of the ordinate and the high block of the cubic power. This supplies a close structural antecedent for (44)–(45). The differential scale, uniformizer and actual divisor representative must still be specified when comparing that polynomial with a Miller coefficient.

Semaev (1998, Lemmas 1–3) gives the directly relevant computational specialization: an additive logarithmic-derivative embedding of an order-p subgroup, evaluated by differentiating the function recurrence in $O(\log p)$ field operations. Rück (1999, p. 806) extends the transfer to divisor class groups and explicitly constructs the local-coefficient cocycle for divisor addition. His addition-function differential may have a pole at the expansion point; in genus one the needed coefficient is its Laurent constant. Repeated doubling computes the transfer. Thus both differentiated Miller evaluation and its local-coefficient cocycle interpretation have explicit antecedents. Galbraith (2012, Section 26.4.1) gives a later account of the p-torsion differential map and its coefficient computation.

The anomalous-curve algorithms form the corresponding cryptographic context. Satoh and Araki (1998, with the 1999 correction) use elliptic Fermat quotients, and Smart (1999) uses a p-adic lift, to solve discrete logarithms when $\#E(\mathbb{F}_p) = p$. Blake, Seroussi and Smart (1999, Chapter V, Section 3) explain the lifting approach. Trace -1 reduces to trace one on a quadratic twist. These order-p transfers account for the exceptional setting of Theorem 1; they do not make the general-trace Laurent coefficient an additive discrete-logarithm embedding on an arbitrary elliptic curve. Lemma 10 fixes the normalization relative to $\omega = dw/v$ and the specified two-pole residue divisor, including the factor of two associated with $(P) - (-P)$. Theorem 2 gives an all-trace identity using the rational curve order as the Miller scalar.

The coordinate and algebraic implementation also uses established constructions. The cubic/quartic model change follows Connell (1999, Section 1.2), with its differential scale made explicit here. Splitting the quadratic algebra when a zero divisor appears is a degree-two instance of dynamic evaluation, or the D5 principle (Della Dora et al., 1985). These ingredients remove particular root-search requirements; they do not constitute a new general root-finding algorithm.

A separate computational lineage provides the natural trace-free comparison. Bostan, Gaudry and Schost (2007) accelerate polynomial recurrences through block products. Harvey (2007, Theorem 10) gives the interval-product theorem used by our selected-coefficient adapter. For fixed degree, (27)–(28) give a $\widetilde{O}(\sqrt{p})$-field-operation route to the same single-pole output, including branch poles, with interpolation and denominator preparation charged. The block-product machinery is classical; its conversion to the weighted-sum output is stated and proved here. Its trace-free setup must be

distinguished from the trace-supplied query model and from dense-query polynomial methods.

Harvey and Sutherland (2014, 2016) address another workload, computing Hasse–Witt matrices over many primes by accumulating remainder trees and forests. The rounded-third companion specializes that machinery to a triangular recurrence acquiring prefix and boundary jointly. Asif, Fité and Pentland (2022), with an appendix by A. V. Sutherland, use Cartier–Manin data and supplementary splitting information to reconstruct L-polynomials of generic Picard curves outside a density-zero set of good primes. Their genus-three reconstruction provides a methodological comparison with our genus-one coefficient evaluation; it does not supply the residue-divisor identity. Frobenius on third-kind differentials in local height computations (Balakrishnan & Besser, 2012) and Cartier coefficient calculations in Dwork crystals (Beukers & Vlasenko, 2021) are additional related frameworks, with different outputs and hypotheses.

The arithmetic applications require further input beyond a fast prefix. The quarter-point boundary uses classical Gauss/CM congruences, with the normalization recorded by Cosgrave and Dilcher (2010); truncated central-binomial congruences are also studied by Sun (2011). The rounded-third boundary uses Jacobi-sum methods of van Wamelen (2002) and the split construction considered in Tal's unrefereed preprint (2026), with a prime conductor dividing $p-1$. Our implementation specializes the anchor intervals and verifies reconstruction certificates. It does not inherit the exponent claim for Tal's distinct extension-field construction, or turn a fast prefix into a fast acquisition of the missing boundary coefficient.

An elementary multiplicative analogy is the finite logarithm (Elbaz-Vincent & Gangl, 2002):

$$\ell_{1,p}(X) = \sum_{j=1}^{p-1} \frac{X^j}{j} \equiv \frac{1 - X^p - (1-X)^p}{p} \pmod{p}. \tag{5}$$

The quotient is formed in $\mathbb{Z}[X]$ before reduction. At a base-field argument, exponentiation modulo p^2 and exact division by p evaluate it with $O(\log p)$ ring operations. This is a comparison with short arithmetic evaluation of a long truncation, without identifying our regular elliptic coefficient with a multiplicative Fermat quotient.

The mathematical focus here is the explicit connection between coefficient sums, normalized logarithmic scalars and rationally weighted character sums, including the cubic/quartic adapters, complete excluded domains, equivalent elliptic-point sums and separately charged field and bit costs. Theorem 1, Lemma 10 and Theorem 2 specify those connections and their normalizations. We do not claim novelty for Miller's short-chain construction, the p-torsion logarithmic transfer or Rück's local-coefficient cocycle. We have not located the complete general-trace scalar identity with its weighted-sum adapters in the sources examined; this bounded comparison establishes neither historical priority nor prior publication of that combined evaluator.

2.1 Notation

The quadratic character is $\chi : \mathbb{F}_p \to \{0, \pm 1\}$, with $\chi(0) = 0$, and $\varepsilon_f(x) = \chi(f(x))$. A single character value is denoted e. Frobenius on points is π; coefficient conjugation is σ; Verschiebung is Ver; and absolute Cartier is $\mathcal{C}$. High-block polynomials are denoted J_f, J_W, so they cannot be confused with an isogeny. The integer curve order is N. The specified residue divisor is $\mathfrak{D}$, whereas Ω denotes an included affine domain. The local Miller coefficient is $\alpha_m(P)$, the differential scale is $\kappa_\omega = -2$, and the addition-function coefficient is $c(S,T)$. Local coordinates and the cutoff indices are defined where used.

3 Local coefficients and the uniform curve order

All differential calculations are in characteristic p. We use the Cartier convention and finite-field power sums

$$\mathcal{C}\left(\sum_n b_n z^n\,dz\right)=\sum_j b_{pj+p-1}^{1/p} z^j\,dz, \qquad \sum_{x\in\mathbb{F}_p} x^k=\begin{cases}-1 & k>0,\ (p-1)\mid k,\\ 0 & \text{otherwise,}\end{cases} \tag{6}$$

where the term for $k=0$ is also zero in $\mathbb{F}_p$. Cartier satisfies $\mathcal{C}(a^p\eta)=a\mathcal{C}(\eta)$ and fixes logarithmic differentials (Cartier, 1957; Serre, 1958).

3.1 The Miller coefficient and its recurrence

For a point R on the short model, use Miller functions and their local constants

$$\begin{aligned}\operatorname{div}(f_{m,R}) &= m(R)-([m]R)-(m-1)(O),\\ \alpha_m(R) &= [z^0]\,\tfrac{\operatorname{dlog} f_{m,R}}{dz}, \quad z=-X/Y.\end{aligned} \tag{7}$$

This is a Laurent coefficient of df/f, not an analytic logarithm, and multiplication of a Miller function by a scalar does not change it. The divisor recurrence is standard (Miller, 2004; Galbraith, 2012, Section 26.3.1). A nonvertical chord or tangent of slope ρ and a vertical line have expansions

$$\begin{aligned}Y-\rho X-\nu &= -z^{-3}\left(1+\rho z+O\left(z^3\right)\right),\\ X-c &= z^{-2}\left(1-cz^2+O\left(z^4\right)\right).\end{aligned} \tag{8}$$

The constant logarithmic derivative is therefore $+\rho$ for the nonvertical line and 0 for the vertical line. Binary doubling and addition update the accumulator by

$$\alpha\leftarrow 2\alpha+\rho, \qquad \alpha\leftarrow\alpha+\rho, \tag{9}$$

respectively. Vertical and identity steps contribute zero. Doubling the accumulator is still necessary when the current point is O.

Algorithm 1. Base-field differentiated Miller chain. Given a point R, scalar $m\geq 1$ and short coefficient A, the following accumulator returns $\alpha_m(R)$ and $[m]R$. The loop reads binary digits without storing intermediate point states.

```
T := R; alpha := 0
for bit in the binary expansion of m, after its leading 1:
    (T, rho) := ADD(T, T)
    alpha := 2*alpha + rho
    if bit == 1:
        (T, rho) := ADD(T, R)
        alpha := alpha + rho
return alpha, T
```

ADD returns $(Q,0)$ for an identity first operand and $(T,0)$ for an identity second operand. Equal abscissae with opposite ordinates give $(O,0)$, including doubling a point of order two. Otherwise the slope is $\rho=(3X_1^2+A)/(2Y_1)$ for a tangent and $\rho=(Y_2-Y_1)/(X_2-X_1)$ for a chord. Set $X_3=\rho^2-X_1-X_2$ and $Y_3=\rho(X_1-X_3)-Y_1$. The doubling update of the accumulator is applied even if ADD returns the identity. This convention covers all intermediate identity and vertical steps.

3.2 The curve order identity

Theorem 2 (Curve order Miller coefficient). Let $p\geq 5$, let $E: y^2=W(x)=x^3+ux+v$ be nonsingular, and put $\omega_0=dx/(2y)$, $t=-x/y$, $N=\#E(\mathbb{F}_p)=p+1-\tau_E$. Write

$$W(x)^{(p-1)/2}=U(x)+H_E x^{p-1}+x^p J_W(x), \qquad \deg U\leq p-2. \tag{10}$$

With the Miller convention (7), for every affine rational point $P = (a, b)$,

$$\alpha_N(P) = bJ_W(a). \tag{11}$$

There is no coprimality or ordinarity hypothesis. Set $\alpha_m(O) = 0$. For $p \nmid N$, the differential has a pole at O, so its Laurent constant is used; it is not its value at O, or a claimed additive embedding of the full rational point group.

Proof. Put $c(S,T) = [t^0](d\log g_{S,T}/dt)$, where $\operatorname{div} g_{S,T} = (S) + (T) - (S+T) - (O)$. A chord or tangent contributes its slope, a vertical line contributes zero, and an identity operand uses $g = 1$, as in (8)–(9). Comparing divisors proves

$$\begin{aligned} \alpha_m(S+T) &= \alpha_m(S) + \alpha_m(T) + c([m]S, [m]T) - mc(S,T), \\ \alpha_{mn}(Q) &= n\alpha_m(Q) + \alpha_n([m]Q). \end{aligned} \tag{12}$$

For the second identity use $f_{mn,Q} \doteq f_{m,Q}^n f_{n,[m]Q}$, where dotted equality allows a nonzero constant. These identities hold for geometric points, including identity endpoints. Negation gives $\alpha_m(-S) = -\alpha_m(S)$. In particular, α_p is additive when one summand is geometric p-torsion.

Suppose first $b \neq 0$. Choose a geometric Q with $[p]Q = P$ and set $\beta = \alpha_p(Q)$. The differential $\vartheta_P = (y+b)\omega_0/(x-a)$ has residue divisor $(P) - (O)$. Since $W(a)^{(p-1)/2} = 1$, division of $W^{(p-1)/2} - 1$ by $x - a$ gives a polynomial of degree below $2p-1$ whose x^{p-1} coefficient is $J_W(a)$. The Cartier coefficient rule, together with $1/y = W^{(p-1)/2}/y^p$, gives $(\mathcal{C} - 1)\vartheta_P = bJ_W(a)\omega_0$. The differential $\theta = -d\log f_{p,Q}$ has the same residues and is Cartier fixed. At O, $\vartheta_P/\omega_0 = -t^{-1} + O(t)$ and $\theta/\omega_0 = -t^{-1} - \beta + O(t)$. Thus $\vartheta_P - \theta = \beta\omega_0$. Applying Cartier and raising to p, with $\mathcal{C}\omega_0 = H_E^{1/p}\omega_0$, proves

$$bJ_W(a) = H_E\beta - \beta^p. \tag{13}$$

Let π be Frobenius and Ver Verschiebung. Since $\pi \operatorname{Ver} = [p]$ and Frobenius is injective on geometric points, $\operatorname{Ver} Q = P$. The trace relation $\pi + \operatorname{Ver} = [\tau_E]$ therefore gives $\pi Q = Q - [N]Q$. Put $T = [N]Q$; then $[p]T = [N]P = O$. Galois compatibility and (12) give

$$\beta^p = \alpha_p(\pi Q) = \beta - \alpha_p(T), \qquad N\beta + \alpha_N(P) = \alpha_{pN}(Q) = \alpha_p(T). \tag{14}$$

The last equality uses the composition identity in both orders and $p = 0$ in the coefficient field. As $H_E \equiv \tau_E \pmod p$, substitution in (13) yields $bJ_W(a) = (H_E - 1 + N)\beta + \alpha_N(P) = \alpha_N(P)$. If $b = 0$, then N is even and $f_{N,P} \doteq (x-a)^{N/2}$; its logarithmic constant is zero, as is the right side of (11). This proves the remaining case. The point Q is a proof device and is never computed.

At trace one, $N = p$, so (11) is the trace-one normalization discussed in Section 5. For the representative $(P) - (-P)$, the function $f_{N,P}/f_{N,-P}$ has logarithmic constant $2\alpha_N(P) = 2bJ_W(a)$. This states explicitly the factor of two for that representative relative to ω_0 and t. When $p \nmid N$, choosing the rational point $Q = [p^{-1} \bmod N]P$ in (13) also gives $bJ_W(a) = (H_E - 1)\alpha_p(Q)$. The theorem avoids this modular inverse and preliminary scalar multiplication. Its annihilator is the same curve order used in Theorem 1 after the appropriate twist.

The local-coefficient cocycle and its short-chain computation have the classical antecedents discussed in Section 2.

4 Weighted sums and the preferred evaluator

Let $p \geq 5$ and let $f \in \mathbb{F}_p[x]$ be any squarefree cubic or quartic, with no condition on its constant term. All weighted sums below are residues in $\mathbb{F}_p$. They differ from integer character sums, incomplete interval sums, and nonlinear sums of the form $\sum_x \chi(\mathcal{R}(x)f(x))$.

Proposition 3 (Polynomial identity at every pole). For a squarefree cubic or quartic over $\mathbb{F}_p$, $p \geq 5$, write $F = f^{(p-1)/2} = T_f + x^p J_f$, $\deg T_f < p$. For every $a \in \mathbb{F}_p$, including branch values and

zero,

$$C_f(a) := \sum_{x \neq a} \frac{\varepsilon_f(x)}{x-a} = -F'(a) - J_f(a). \tag{15}$$

Proof. Put $Q = (F - F(a))/(x-a)$. The reciprocal sum over $x \neq a$ is zero, so $C_f(a) = \sum_x Q(x) - Q(a)$. Here $\deg Q \leq 2p-3$; the power sums (6) give $\sum_x Q(x) = -[x^{p-1}]Q = -J_f(a)$, while $Q(a) = F'(a)$. At a nonbranch point, $F'(a) = -\varepsilon_f(a) f'(a)/(2f(a))$. At a branch point, $F'(a) = 0$, since $(p-1)/2 \geq 2$. For $a \neq 0$, (15) becomes $C_f(a) = (T_f(a) - \varepsilon_f(a))/a - F'(a)$, which will also recover the truncation.

Theorem 4 (Uniform nonbranch query). Let $f(a) \neq 0$, set $e = \varepsilon_f(a)$, and put

$$g(z) = \frac{f(a+z)}{f(a)} = 1 + g_1 z + g_2 z^2 + g_3 z^3 + g_4 z^4, \quad g_1 = \frac{f'(a)}{f(a)}, \quad \widetilde{\tau} = e\tau_f, \quad N = p + 1 - \widetilde{\tau}. \tag{16}$$

Here τ_f is the exact projective trace and $g_4 = 0$ for a cubic. Apply the fixed-degree model change in Appendix A to $v^2 = g(z)$, taking $(0,1)$ as origin, and let R be the image of $(0,-1)$. On the resulting short model,

$$\boxed{C_f(a) = e\left(-\frac{g_1}{4}(N-2) - \frac{\alpha_N(R)}{2}\right).} \tag{17}$$

One base-field differentiated Miller chain evaluates this expression; it requires neither a square root nor a quadratic algebra.

Proof. Translation and normalization give $\varepsilon_f(a+z) = e\varepsilon_g(z)$ and $C_f(a) = eC_g(0)$. If $N_\infty(f)$ is the number of rational points at infinity, then $N_\infty(f) = 1$ for a cubic and $1 + \chi(\ell_f)$ for a quartic with leading coefficient ℓ_f. Fibre counting gives $\tau_f = 1 - N_\infty(f) - \sum_x \varepsilon_f(x)$. For a quartic normalization also changes the infinity character by e; this identity proves the signed integer relation $\tau_g = e\tau_f$ in both degrees.

The model change in Appendix A has short coefficients and point

$$\begin{aligned} A_g &= -4g_4 + g_1 g_3 - g_2^2/3, \\ B_g &= g_3^2 - 8g_2 g_4/3 + g_1^2 g_4 - g_1 g_2 g_3/3 + 2g_2^3/27, \\ R = (x_R, y_R) &= (g_1^2/4 - 2g_2/3, -g_3 + g_1 g_2/2 - g_1^3/8). \end{aligned} \tag{18}$$

The formulas extend to an isomorphism of smooth projective curves. Substitution, or expansion at $(0,-1)$ followed by (55), gives R. An exact identity of rational functions is

$$\frac{1}{z} = \frac{1}{2}\varphi_R - \frac{g_1}{4}, \qquad \varphi_R = \frac{Y + y_R}{X - x_R}. \tag{19}$$

It follows directly from (55) and does not require division by y_R. Apparent singularities at removable points use their regular values. When $y_R \neq 0$, the value at $-R$ is $-W_g'(x_R)/(2y_R)$, where $W_g = X^3 + A_g X + B_g$. Summing the other fibres and applying (15) on the short model gives

$$\sum_{\substack{S \in E_g(\mathbb{F}_p) \\ S \notin \{O,R\}}} \varphi_R(S) = y_R C_{W_g}(x_R) - \frac{W_g'(x_R)}{2y_R} = -y_R J_{W_g}(x_R) = -\alpha_N(R). \tag{20}$$

If $y_R = 0$, negation pairs the terms $Y/(X - x_R)$; the fixed points contribute zero, and Theorem 2 gives $\alpha_N(R) = 0$. Thus the same identity holds for a nonzero rational point of order two.

The function $1/z$ vanishes at every point at infinity of the cubic or quartic model. Its affine point sum with $z \neq 0$ is $C_g(0)$, since $\sum_{z \neq 0} 1/z = 0$. The short-model sum excludes exactly O, R, hence has $N - 2$ terms. Summing (19) and using (20) proves (17). The geometric construction in Theorem 2 supplies its sign for all traces; no exceptional constant is selected experimentally.

In particular, a zero pole and a zero constant term of the original f present no normalization obstruction when $f(a) \neq 0$. Every pole is used as its own normalization origin. The same cached original trace supplies each twist trace. Formula (17) includes $p = 5$, even when $N = 2p$; for example it returns $C_{x^3+3x}(1) = 1$ over $\mathbb{F}_5$, with $\tau_f = -4$. This extends the original $p \geq 7$ formulation. Branch poles remain outside the fast query.

4.1 Recovering the coefficient sum

Corollary 5 (One chain evaluation of the truncation). Let $p \geq 5$, let f be a squarefree cubic or quartic, and define $T_f = f^{(p-1)/2} \bmod x^p$. For $\mu \neq 0$ with $f(\mu) \neq 0$, use the data $e = \varepsilon_f(\mu)$, $N = p + 1 - e\tau_f$ and R of Theorem 4. Then

$$T_f(\mu) = e + \mu C_f(\mu) - \frac{e\mu f'(\mu)}{2f(\mu)} = e\left(1 - \frac{\mu N f'(\mu)}{4f(\mu)} - \frac{\mu}{2}\alpha_N(R)\right). \tag{21}$$

Consequently this single base-field chain evaluates $S_f(\lambda)$ at $\lambda = \mu^{-1}$. No nonzero constant term is needed for this computational statement.

Proof. The first equality is Proposition 3 with $F'(\mu) = -ef'(\mu)/(2f(\mu))$. Substitution of (17) combines the two derivative terms into $-e\mu N f'(\mu)/(4f(\mu))$. The equality $S_f(\lambda) = T_f(\mu)$ follows from $\lambda^{p-1} = 1$, without a normalization at zero. At $\mu = 0$, direct evaluation gives $T_f(0) = f(0)^{(p-1)/2}$; at $\lambda = 0$, use the Hasse coefficient. This corollary is a computational extension of the coefficient query and does not alter the $p \geq 7$ hypotheses of Theorem 1.

For the normalized data of Theorem 1, its two-pole scalar and the twist-point scalar therefore satisfy

$$r\Psi_{M_e}(\mathfrak{D}) = eH - 1 + \frac{\mu N f'(\mu)}{4f(\mu)} + \frac{\mu}{2}\alpha_N(R), \qquad N = M_e. \tag{22}$$

Here R lies on the normalized twist, whereas $\mathfrak{D}$ is the specified divisor on the original model. The differential conventions in the two constructions are essential to this equality. It identifies the same coefficient output in the structural and computational formulations.

4.2 Several poles and complete excluded domains

Theorem 6 (Supplied simple pole weights). Let $\mathcal{A} = \{a_1, \ldots, a_s\} \subset \mathbb{F}_p$ be distinct, $\Omega = \{x \in \mathbb{F}_p : x \notin \mathcal{A}\}$, and let $\mathcal{R}(x) = c + \sum_i r_i/(x - a_i)$ be supplied in partial fractions. Zero residues are allowed, so $\mathcal{A}$ records the entire excluded set, including extra exclusions. Put $w_i = \varepsilon_f(a_i)$, $K_f = \sum_x \varepsilon_f(x) = 1 - N_\infty(f) - \tau_f$, and $D_i(v) = \sum_{j \neq i} v_j/(a_i - a_j)$. Then

$$\begin{aligned}
\sum_{x \in \Omega} \mathcal{R}(x)\varepsilon_f(x) &= c\left(K_f - \sum_i w_i\right) + \sum_i r_i\big(C_f(a_i) + D_i(w)\big), \\
\sum_{\substack{P \text{ affine} \\ x(P) \in \Omega}} \mathcal{R}(x(P)) &= c\left(K_f - s - \sum_i w_i\right) + \sum_i r_i\big(C_f(a_i) + D_i(1 + w)\big).
\end{aligned} \tag{23}$$

The difference of the point sum and character sum is the elementary baseline $-sc + \sum_i r_i D_i(1)$. For one reciprocal pole and $c = 0$, it is zero. The two sums therefore express the same evaluation problem up to an explicitly computable baseline.

Proof. The oracle $C_f(a_i)$ excludes only a_i. Removing each other a_j subtracts $w_j/(a_j - a_i)$, giving the plus correction $D_i(w)$. The affine fibre over x has the integer size $1 + \varepsilon_f(x)$. Its reciprocal baseline is $\sum_{x \in \Omega}(x - a_i)^{-1} = D_i(1)$; its constant baseline is $|\Omega| = p - s \equiv -s \pmod p$. These facts prove both formulas and their difference. Corrections must use all of $\mathcal{A}$, even where $r_i = 0$. The formulas also hold for Ω empty.

The weight tends to c at infinity. The projective rational-point sum, retaining all rational points at infinity, consequently adds $cN_\infty(f)$ to the affine sum. No character value at infinity is assigned. A positive-degree polynomial part instead has a pole there and needs a separately specified domain; a quartic with $N_\infty(f) = 0$ has no rational infinity point to exclude. For a general function $A(x) + yB(x)$, cancellation of the odd part by negation is valid only on whole included fibres where the function is regular, with removable singularities evaluated by their limits.

4.3 Query costs and admissibility

Corollary 7 (Field and bit bounds). Assume every active pole is nonbranch and the exact signed trace is supplied. With $n = \lceil \log_2 p \rceil$, Theorem 4 uses $O(n)$ field operations, including $O(n)$ inversions, and a constant number of field registers with $O(n)$ control bits. Writing $M(n), I(n)$ for multiplication and inversion bit costs, a safe query bound is $O(n(M(n) + I(n) + n))$. This is $O(n^3)$ classically and $\widetilde{O}(n^2)$ with fast arithmetic. Trace acquisition and primality certification are separate input costs.

Here a field operation includes inversion; the number of inversions is also recorded separately. More precisely, $M(n)$ bounds multiplication of two n-bit integers together with reduction modulo p, and $I(n)$ bounds inversion of a nonzero residue modulo p. Addition and input/output of one residue cost $O(n)$ bits. Schoolbook arithmetic and classical extended Euclid allow $M(n), I(n) = O(n^2)$. Fast multiplication and half-gcd allow $M(n) = \widetilde{O}(n)$ and $I(n) = O(M(n)\log(n+1))$. These choices justify the two bit bounds above; they do not identify field operations with bit operations.

For (23) with $s \geq 1$, character evaluation at all exclusions and schoolbook corrections give $O(sn + s^2)$ field operations. Batch inversion of the nonzero differences gives the bit bound

$$O(sn(M(n) + I(n) + n) + s^2 M(n) + I(n)), \quad O(sn^3 + s^2 n^2) \text{ classically.} \tag{24}$$

An exclusion with zero residue may be a branch point, since its oracle value is not needed. No denominator factorization or unsupplied partial-fraction decomposition is included in these bounds.

Proof. Forming (16), (18) costs a constant number of field operations apart from the character exponentiation. By Hasse's bound $N = O(p)$, so the binary Miller chain has $O(n)$ steps. All denominators in the fixed-degree model change are units for $p \geq 5$. Identity and vertical steps are covered by (8)–(9), including an intermediate endpoint O. Multiplying s query costs and adding the pairwise corrections proves (24). If $s = 0$, only the trace-derived constant term remains. Reading s arbitrary field inputs already costs $\Omega(sn)$ bits. The endpoint check from the chain is a necessary consistency check for a supplied trace, not a trace certificate.

For completeness, the correction vector has a fast polynomial construction. Set $B(X) = \prod_j (X - a_j)$ and $W_v(X) = \sum_j v_j B(X)/(X - a_j)$. Taylor expansion at a_i gives

$$D_i(v) = \frac{W_v'(a_i)}{B'(a_i)} - \frac{v_i B''(a_i)}{2B'(a_i)}. \tag{25}$$

Indeed the derivative of the ith summand is $v_i B''(a_i)/2$, and the remaining derivatives sum to $B'(a_i)D_i(v)$. A product tree constructs B and W_v: at a leaf use $(X - a_i, v_i)$, and combine disjoint children by $(B_L B_R, W_L B_R + W_R B_L)$. Derivative formation, remainder-tree evaluation and one batch inversion then cost $O(\mathsf{M}_{mpol}(s)\log(s+1))$ field operations under the usual polynomial-arithmetic cost assumptions, plus one scalar inversion. This proved polynomial construction is not implemented in the supplied pairwise adapter. It improves the correction term in (24) accordingly, with storage for the polynomial trees. It is an arithmetic bound, not a new benchmark. For fixed curve and excluded set, caching the oracle values and correction vectors makes subsequent coefficient-only changes $O(s)$-operation dot products.

The supplied representation contains $\Theta(s)$ residues and pole coordinates, so reading it takes $O(sn)$ bits. Distinctness validation can use the same nonzero pairwise differences as the correction routine; output formation takes $O(s)$ field operations and $O(n)$ bits for one final residue. Optional decomposition of an unprocessed rational function is a separate task depending on numerator and denominator degrees and on factorization; no bound in s, n alone is assigned to it. With fixed p, f, a new pole set reuses the trace but requires its own oracle calls and exclusion corrections. With fixed $p, f, \mathcal{A}$, only the coefficients need be read again and the cached linear forms cost $O(s)$ field operations, or $O(s(M(n) + n))$ bits. A change of prime or an unrelated defining polynomial requires fresh trace preparation; a proved coordinate change or quadratic twist transports a known trace by the stated sign rule.

The pairwise implementation uses one batch inversion and $O(s^2)$ temporary field elements; its persistent prepared state uses $O(s)$ field elements. The polynomial-tree construction has its own tree storage. These results establish an accelerated class of constant-plus-simple-pole weights. They do not assert the same bound for every rational weight. At a branch pole (15) remains valid, but its additional high coefficient has not been supplied by Theorem 4. Higher-order poles require further local coefficients or Hasse derivatives. A base-field pointwise identity such as (4) cannot be differentiated in a transcendental argument to obtain them. The trace-free selected-coefficient method below remains available at branch poles. Likewise, the extension-field and higher-genus problems in Section 8 require separate proofs.

Corollary 8 (Deterministic preparation and evaluation). For prime $p \geq 5$, squarefree cubic or quartic f, and nonzero λ with $f(\lambda^{-1}) \neq 0$, the coefficient query is deterministically computable in time polynomial in $\log p$. With N_{qry} queries on one prepared curve, classical arithmetic gives

$$T_{\mathrm{total}} = T_{\mathrm{Schoof}}(p) + O(N_{\mathrm{qry}}(\log p)^3), \qquad T_{\mathrm{Schoof}}(p) = O((\log p)^9). \tag{26}$$

Proof. Fixed-degree squarefreeness tests and model changes take polynomial time. A nonbranch pole supplies the normalized model with origin $(0, 1)$. More generally, a nonroot of f is found among $\deg f + 1$ fixed distinct base-field elements; normalization there gives a short model with a rational origin, without a square-root search. Schoof's deterministic point count supplies its exact trace, and the known character sign recovers the original trace (Schoof, 1985, 1995). Corollary 5 and Corollary 7 then apply. At $\lambda = 0$, the Hasse coefficient is the exact trace reduced modulo p, including the quartic infinity correction. The displayed preparation bound is a conservative bound for Schoof's original algorithm. The query uses a constant number of field registers and $O(\log p)$ control bits; this excludes point-counting workspace.

Primality certification is separately charged unless a prime is supplied. Practical trace preparation may use SEA or an exact routine such as PARI/GP's `ellap`; the latter's available data and algorithm affect the cost (PARI Group, n.d.). The historical character-sum experiment used PARI without SEA data and did not benchmark Schoof. Repeated poles on one curve reuse the original trace, since the signed trace of each normalized twist is $e\tau_f$.

4.4 A trace-free selected-coefficient comparator

Put $d = \deg f$, $h = (p-1)/2$, $Q_a(z) = z^d f(a + 1/z)$, and $k = dh - p < p$. Since $(a+z)^p = a + z^p$ and $\deg J_f < p$, the coefficient of z^p in $F(a+z) = T_f(a+z) + (a+z^p)J_f(a+z)$ is $J_f(a)$. Reversing this polynomial proves

$$J_f(a) = [z^k]Q_a(z)^h, \qquad Q_a(0) = \ell_f \neq 0. \tag{27}$$

This includes zero and branch poles without trace preparation. If $Q_a = \sum_{j=0}^{d} g_j z^j$ and $Q_a^h = \sum t_k z^k$, then

$$k g_0 t_k = \sum_{j=1}^{\min(d,k)} ((h+1)j - k) g_j t_{k-j}, \qquad t_0 = g_0^h. \tag{28}$$

All denominators are nonzero for $1 \leq k < p$. The state $k! g_0^k (t_k, \ldots, t_{k-d+1})$ evolves through a fixed-size matrix linear in k: its top row is $((h+1)j - k)g_j$, and its subdiagonal entries are kg_0. Negative-index coefficients are zero. Streaming costs $O(p)$ field operations and constant storage.

Bostan–Gaudry–Schost block products (Bostan et al., 2007) and Harvey's interval-product theorem (Harvey, 2007, Theorem 10) give $\widetilde{O}(\sqrt{p})$ field operations for fixed d, with $\widetilde{O}(\sqrt{p})$ storage. Harvey's theorem concerns linear matrix polynomials, ordered interval products and suitable invertible interpolation factors. For the single interval $[1, k]$, its sufficient unit condition holds because, with $s_0 = \lfloor \log_4 k \rfloor$, $2^{s_0} + 1 < p$. The interpolation product and its inverse, and the scalar denominator $k! g_0^k$, must be acquired within the same square-root budget; they are not free preparation. Handle $k \leq 1$ directly. The precise ring-operation bound is $O(\mathsf{MM}(d)\sqrt{k} + d^2 \mathsf{M}_{\mathrm{pol}}(\sqrt{k}))$, apart from the smaller initialization exponentiation. The interval product order must agree with the column-state recurrence.

The literature supplies the block-product machinery; (27) supplies the matched weighted-sum adapter. These trace-free costs and the trace-supplied cost in Corollary 7 describe different preparation models. Neither is an unconditional ranking for dense-query workloads.

5 The residue divisor interpretation

All differential identities are in characteristic p. Frobenius π on points means the p-power map on coordinates. Coefficient conjugation σ raises constants to their pth powers while leaving coordinate variables fixed. Put

$$\omega = \frac{dw}{v}, \qquad \theta_\lambda = \frac{dw}{(1-\lambda w)\, v}. \tag{29}$$

The differential ω is holomorphic and nowhere zero. For nonzero admissible λ, the only poles of θ_λ are simple poles at Q_+ and Q_- with residues $-r$ and r.

Since $f(0) = 1$, w is a uniformiser at O. The relation $v^{p-1} = f^h$ gives the coefficient selection directly:

$$\begin{aligned} \theta_\lambda &= v^{-p} \tfrac{f^h}{1-\lambda w}\, dw, \\ \mathcal{C}(\omega) = H\omega, \qquad & \left(\tfrac{\mathcal{C}(\theta_\lambda)}{\omega}\right)(O) = S_f(\lambda)^{1/p}. \end{aligned} \tag{30}$$

Here $H \in \mathbb{F}_p$. The trace congruence follows from the finite-field power sums (6):

For a cubic, only c_{p-1} survives in the sum of $f(w)^h$, and the projective curve has one point at infinity. For a quartic, $c_{2p-2} = f_4^h = (f_4/p)$ also survives; the infinity contribution is $1 + (f_4/p)$ and cancels this extra term. In both cases $\tau \equiv H \pmod p$.

Proposition 9 (Regular residue identity). Let $\lambda \neq 0$ be in any perfect extension of $\mathbb{F}_p$, with $f(1/\lambda) \neq 0$. Suppose $M \geq 1$, $p \nmid M$, and fix $F_{\mathfrak{D}}$ with $\operatorname{div}(F_{\mathfrak{D}}) = M\mathfrak{D}$ as divisors. Then

$$M\left(S_f(\lambda) - H\right) = r\left(H\Psi_M(\mathfrak{D}) - \Psi_M(\mathfrak{D})^p\right). \tag{31}$$

Proof. The residues cancel in $M\theta_\lambda + r\, dF_{\mathfrak{D}}/F_{\mathfrak{D}}$, so this differential is $c\omega$ for some scalar c. Evaluation at O gives $c = M + r\Psi_M(\mathfrak{D})$. Apply Cartier, evaluate at O, and raise to the pth power to obtain

$$MS_f(\lambda) + r\Psi_M(\mathfrak{D})^p = \left(M + r\Psi_M(\mathfrak{D})\right) H. \tag{32}$$

This is (31). For the fixed divisor $\mathfrak{D}$, replacing $F_{\mathfrak{D}}$ by $F_{\mathfrak{D}}^k$ gives $\Psi_{kM}(\mathfrak{D}) = k\Psi_M(\mathfrak{D})$ in characteristic p.

For a base-field argument, $\pi(Q_+) = Q_e$, so $\pi(\mathfrak{D}) = e\mathfrak{D}$ as divisors. Frobenius's characteristic equation annihilates its divisor class by $M_e = p + 1 - e\tau$. Since $F_{\mathfrak{D}}^\sigma$ differs from $F_{\mathfrak{D}}^e$ by a constant, $\Psi_{M_e}(\mathfrak{D})^p = e\Psi_{M_e}(\mathfrak{D})$. Substitution into (31), using $M_e \equiv 1 - eH \pmod p$, proves (4) when $p \nmid M_e$. Changing the sign of v_0 negates both r and $\Psi_{M_e}(\mathfrak{D})$.

Hasse's bound gives $|\tau| \leq 2\sqrt{p} < p - 1$ for $p \geq 7$. Thus $p \mid M_e$ occurs precisely when $\tau = e$, and then $M_e = p$. The class of $\mathfrak{D}$ is nonzero: distinct points on a genus-one curve cannot have principal difference. It therefore has order p. The regular identity loses its scalar in this case. The bound excludes $p = 5$, where traces ± 4 are possible, from this particular degeneracy argument.

5.1 Direct exceptional normalization

Assume $\tau = e \in \{1, -1\}$ and identify $E = (v^2 = f(w), P_+)$, so $O = P_+$ and $\omega = dw/v$. Work over $k = \overline{\mathbb{F}}_p$. For a degree-zero divisor $\mathfrak{D}_0$ with p-torsion class, choose $\operatorname{div}(F_{\mathfrak{D}_0}) = p\mathfrak{D}_0$. Its logarithmic differential is holomorphic and defines

$$dF_{\mathfrak{D}_0}/F_{\mathfrak{D}_0} = \Psi_p([\mathfrak{D}_0])\,\omega, \qquad \Psi_p : \operatorname{Pic}^0(E)[p](k) \to k. \tag{33}$$

A principal change $\mathfrak{D}_0 \mapsto \mathfrak{D}_0 + \operatorname{div}(g)$ changes the function by a constant times g^p and leaves its logarithmic differential unchanged. Multiplication of functions gives additivity. The map is injective by Serre's logarithmic-differential construction (Serre, 1958, Proposition 10). Since $\mathcal{C}\omega = e\omega$, its image lies in the one-dimensional $\mathbb{F}_p$-space

$$\mathscr{L}_e = \{b \in k : b^p = eb\}. \tag{34}$$

By the residue divisor of a differential we mean the formal sum of its residues at geometric points. When compared with an integer divisor, its coefficients are reduced in k, hence modulo p. In particular, logarithmic residues are integer multiplicities reduced modulo p.

Separable pullback. Cartier commutes with pullback by a separable morphism of smooth curves. Indeed, in a separating coordinate x, a rational differential has a unique expression $\sum_{i=0}^{p-1} u_i^p x^i \, dx$, and Cartier selects $u_{p-1}\, dx$. A separable pullback keeps the coordinate separating, so applying the same expression after pullback proves the assertion.

We use the kernel differential and quasi-period construction in Voloch (1997): Θ_R in (38) corresponds to his differential $f_P\omega$, and (39) is its translation law. The proof below fixes the scale and residue divisor for both signs. The published discussion after Voloch's Proposition 3 prints a negative sign in the quasi-period/logarithmic-derivative formula. With $t_T(Q) = Q + T$ and divisor $p((T) - (O))$, the divisor calculation in (37) gives the positive sign used here; the proof determines that sign directly.

Lemma 10 (Exceptional Cartier defect). Suppose $\tau = e \in \{1, -1\}$. Let $\mathfrak{D}_0$ be a degree-zero integer divisor whose class is killed by p and whose coefficient conjugate is $\sigma(\mathfrak{D}_0) = e\mathfrak{D}_0$ as divisors. If $\theta_{\mathfrak{D}_0}$ is defined over $\mathbb{F}_{p^2}$, has only simple poles with residue divisor $\mathfrak{D}_0$, and satisfies $\sigma(\theta_{\mathfrak{D}_0}) = e\theta_{\mathfrak{D}_0}$, then

$$(\mathcal{C} - \mathrm{id})\, \theta_{\mathfrak{D}_0} = \Psi_p([\mathfrak{D}_0])\, \omega. \tag{35}$$

Equivalently, the Cartier-defect map and the logarithmic transfer are the same additive map from geometric p-torsion into $\mathscr{L}_e$; the normalization factor is exactly 1.

Proof. First consider $\mathfrak{D}_P = (P) - (O)$ for $e = 1$, or $\mathfrak{D}_P = (P) - (-P)$ for $e = -1$, where $P \neq O$ is geometric p-torsion. Let Ver be Verschiebung, identifying the Frobenius twist with E through its model over $\mathbb{F}_p$. The Frobenius relations and invariant differential give

$$\pi + \mathrm{Ver} = [e], \qquad \pi\,\mathrm{Ver} = [p], \qquad \mathrm{Ver}^*\,\omega = e\omega, \qquad \mathcal{C}\omega = e\omega. \tag{36}$$

The curve is ordinary, so Ver is separable and étale of degree p. Its kernel $K = \ker \mathrm{Ver}(k)$ is the cyclic group of geometric p-torsion, and every $T \in K$ automatically satisfies $\pi(T) = eT$. Thus $K = E(\mathbb{F}_p)$ if $e = 1$; if $e = -1$, it is the group $\{T : \pi(T) = -T\}$ corresponding to the rational points on the anomalous quadratic twist.

The divisor $\Delta = \sum_{T \in K} (T) - p(O)$ is principal: its group sum is O, because nonzero points pair as $T, -T$. It is Galois stable. The one-dimensional Riemann–Roch space $L(-\Delta)$ is defined over $\mathbb{F}_p$, so it has a nonzero base-field element s with $\operatorname{div}(s) = \Delta$. Put $\xi = \operatorname{dlog} s$ and $\psi(T) = \Psi_p([(T) - (O)])$. With translation $t_T(Q) = Q + T$, permutation of the kernel points gives

$$\operatorname{div} \frac{t_T^* s}{s} = p(O) - p(-T), \qquad t_T^*\xi - \xi = \psi(T)\omega. \tag{37}$$

The sign is positive because $[(O) - (-T)] = [(T) - (O)]$. Also $\mathcal{C}\xi = \xi$ and $\psi(T)^p = e\psi(T)$.

For $R \in E(k)$, put $Q = \mathrm{Ver}(R)$. Equation (37) makes $t_{-R}^*\xi - \xi$ invariant under K. Division by $\mathrm{Ver}^*\,\omega$ gives a K-invariant rational function, which descends through the separable quotient Ver. Hence there is a unique rational differential Θ_R on the target with

$$\begin{aligned} \mathrm{Ver}^*\,\Theta_R &= t_{-R}^*\xi - \xi = \operatorname{dlog} \tfrac{t_{-R}^* s}{s}, \\ \operatorname{div} \tfrac{t_{-R}^* s}{s} &= \mathrm{Ver}^*\,((Q) - (O)) - p\,((R) - (O)). \end{aligned} \tag{38}$$

As Ver is étale, Θ_R has only simple poles and residue divisor $(Q)-(O)$; the terms divisible by p have zero logarithmic residues. The separable-pullback observation and injectivity of separable pullback give $\mathcal{C}\Theta_R = \Theta_R$. Translation and coefficient conjugation give

$$\Theta_{R+T} = \Theta_R - e\psi(T)\omega \quad (T \in K), \qquad \sigma(\Theta_R) = \Theta_{\pi(R)}. \tag{39}$$

Choose R with $\mathrm{Ver}(R) = P$, so $\pi(R) = eR - P$. For $e = 1$, set $\zeta = \Theta_R$ and $\beta = \psi(P)$; for $e = -1$, set $\zeta = \Theta_R - \Theta_{-R}$ and $\beta = 2\psi(P)$. In each case ζ has residue divisor $\mathfrak{D}_P$ and $\beta = \Psi_p([\mathfrak{D}_P])$. The factor 2 is required because $[(P)-(-P)] = 2P$. Equation (39) gives

$$\mathcal{C}\zeta = \zeta, \qquad \sigma(\zeta) = e\zeta + e\beta\omega, \qquad \beta^p = e\beta. \tag{40}$$

In the negative case, explicitly, $\sigma(\zeta) = \Theta_{-R-P} - \Theta_{R+P} = -\zeta - 2\psi(P)\omega$. A prescribed eigen differential θ_P differs from ζ by $c\omega$. Its eigencondition gives

$$c^p - ec = -e\beta, \qquad ec^{1/p} - c = \beta. \tag{41}$$

Thus $(\mathcal{C} - \mathrm{id})\,\theta_P = \beta\omega$.

To pass to $\mathfrak{D}_0$, represent its class by $\mathfrak{D}_P$, choosing half the class when $e = -1$; multiplication by 2 is invertible on the group of order p. The zero class is handled by $\mathrm{dlog}\, g$ directly. Choose a third-kind differential η with residues $\mathfrak{D}_P$ over $\mathbb{F}_{p^2}$ and replace it by $(\eta + e\sigma\eta)/2$ to satisfy the eigencondition. Conversely, the eigencondition implies descent to $\mathbb{F}_{p^2}$ because $\sigma^2\theta = \theta$. Write $\mathfrak{D}_0 - \mathfrak{D}_P = \mathrm{div}(g)$. Its eigencondition implies $g^\sigma = cg^e$ for some $c \in k^\times$, so $\mathrm{dlog}\, g$ is an eigen differential. The difference $\theta_{\mathfrak{D}_0} - \theta_P - \mathrm{dlog}\, g = b\omega$ has $b^p = eb$, hence is Cartier fixed. Cartier also fixes $\mathrm{dlog}\, g$. This proves (35).

This reduction shows that the defect depends only on the divisor class: changing an admissible differential adds a Cartier-fixed $b\omega$, and a principal eigen divisor contributes $\mathrm{dlog}\, g$, also Cartier fixed. The map is additive. Its image lies in $\mathscr{L}_e$, as does the injective transfer (33). The proof determines their comparison scalar as 1, including the factor of 2 in the negative eigencase. The kernel function and the preimage under Ver are proof constructions and are not computed by the evaluator.

Completion of Theorem 1. In the exceptional case put $\widehat{\theta} = -\theta_\lambda/r$. It has residue divisor $\mathfrak{D}$ and coefficient eigenvalue e. Lemma 10 gives $(\mathcal{C} - \mathrm{id})\,\widehat{\theta} = \Psi_p(\mathfrak{D})\omega$. Since $r^{1/p} = er$, evaluate at O to obtain

$$S_f(\lambda)^{1/p} = e - er\Psi_p(\mathfrak{D}), \qquad (er\Psi_p(\mathfrak{D}))^p = er\Psi_p(\mathfrak{D}). \tag{42}$$

Raising to the pth power proves (4) with $H = e$.

For concrete exceptional examples, direct expansion and the normalized Miller evaluator give the following values in $\mathbb{F}_{11}$, with $\lambda = 1$ and origin $(0, 1)$:

Table 1. Exceptional examples over $\mathbb{F}_{11}$.

$f(w)$	τ	$d = f(1)$	e	H	$S_f(1)$	$r\Psi_p(\mathfrak{D})$
$1 + w + 3w^3$	1	5	1	1	9	3
$1 + w + 2w^2 + 3w^3$	-1	7	-1	10	0	1

5.2 The two logarithmic scalars

With the data of Theorem 1, Proposition 3 also gives the original character-sum form

$$C_f(\mu) = \frac{H - e}{\mu} + \frac{ef'(\mu)}{2f(\mu)} - \frac{e}{v_0}\Psi_{M_e}(\mathfrak{D}). \tag{43}$$

This identity is now a consequence of the coefficient formula, while (22) compares its scalar directly with the preferred twist-point chain. The residue-divisor normalization and the base-field evaluator thus have distinct roles.

On a trace-one short model, Theorem 2 specializes to the classical form

$$W^h = U + x^{p-1} + x^p J_W, \qquad yJ_W(x) = \Psi_p([(P) - (O)]), \quad P = (x, y), \tag{44}$$

relative to $\omega_0 = dx/(2y)$. The representative $(P) - (-P)$ has twice this scalar. A trace-minus-one curve becomes trace one on its quadratic twist; the negative Frobenius eigenspace becomes the rational eigenspace of that twist. This explains the relation between the two signs, while Lemma 10 proves the normalization for the actual two-pole divisor.

The Cartier comparison used in Theorem 2, valid for a nonbranch rational point $P = (a, b)$ on any of its short models, is

$$(\mathcal{C} - \mathrm{id}) \left(\frac{y + b}{x - a} \omega_0 \right) = bJ_W(a)\omega_0. \tag{45}$$

6 Application to the original quarter-point sum

For an odd prime p, put $h = (p - 1)/2$ and define the rational sequence

$$t_0 = 1, \qquad t_{i+1} = t_i \frac{h + 2 + i}{2i + 2}, \qquad U_p(L) = \sum_{i=0}^{L-1} (h + 1 + i)\, t_i. \tag{46}$$

Reduction is well defined for $0 \le L \le h$, because the denominators use powers of two and factorials below p. Put $a_i = \binom{2i}{i} 8^{-i}$ and $B_L = \sum_{i<L} a_i$. Upper negation gives, in $\mathbb{F}_p$,

$$t_i = 2^{-i} \binom{h + 1 + i}{i} \equiv 2^{-i} (-1)^i \binom{-3/2}{i} = (2i + 1) \binom{2i}{i} 8^{-i}. \tag{47}$$

The ratio $a_{i+1}/a_i = (2i + 1)/(4(i + 1))$ gives the telescoping certificate

$$\left((2i + 1)^2 - 8 \right) a_i = -4(i + 1)(2i + 7)\, a_{i+1} + 4i(2i + 5)\, a_i. \tag{48}$$

Consequently,

$$U_p(L) \equiv 4B_L - 2L(2L + 5)\, a_L \pmod{p}. \tag{49}$$

Both the prefix and the boundary coefficient are required for this weighted sum. Now take $p = 4n + 1 \ge 13$, so $h = 2n$. Since $\binom{h}{i} (-1/2)^i = a_i$ in $\mathbb{F}_p$, B_n is a strict midpoint truncation of the binomial row of length $h + 1$, although $n = (p - 1)/4$ is a quarter-point relative to p. The full row sum is $(1/2)^h = (2/p)$.

Set $f(w) = 1 - w^4/2$. Its coefficient of w^{4i} in f^h is a_i, so $H = a_n$ and $S_f(1) = B_n + a_n$. Theorem 1 and (49) give

$$B_n = -er\Psi_{M_e}(\mathfrak{D}), \qquad U_p(n) \equiv 4B_n + \frac{9}{4} a_n \pmod{p}, \tag{50}$$

where $e = (2/p)$ and $r = 1/v_0$, $v_0^2 = 1/2$. The curve has $j = 1728$ and rational nonzero two-torsion, so its trace is even and the exceptional traces ± 1 do not occur in this particular application.

The trace is classical complex-multiplication data. With $p = u^2 + s^2$ and the odd coordinate normalized by $u \equiv 1 \pmod 4$, Gauss's congruence is

$$\binom{2n}{n} \equiv 2u \pmod{p}, \qquad a_n \equiv 2u\, 8^{-n} \pmod{p}. \tag{51}$$

This convention agrees with Cosgrave and Dilcher (2010, Theorem 1). For $p \ge 17$, Hasse's bound $2\sqrt{p} < p/2$ makes the trace the representative of a_n in $(-p/2, p/2)$; the remaining prime $p = 13$ has trace 4, checked directly. Cornacchia preparation and exponentiation supply the boundary and trace

for a Miller evaluation of the prefix. The benchmarked preparation includes a sequential nonresidue scan. Corollary 8 gives the deterministic polynomial-time alternative by computing the exact trace, using $a_n = H$, and evaluating $B_n = S_f(1) - H$. At $p = 97$, $n = 24$, $B_n = 33$, $a_n = 79$ and $U_p(n) = 43$.

The congruence literature for truncated central-binomial sums includes Sun (2011). The example uses those classical congruences to supply the boundary. The elliptic-Fermat-quotient interpretation is established in the anomalous setting of Satoh and Araki (1998, 1999); applying that terminology to the regular $\Psi_{M_e}(\mathfrak{D})$ in this example is only an analogy, since it retains a specified divisor and annihilator.

7 Rounded-third prefix and boundary

For primes $p \geq 11$ with $p \equiv 2 \pmod 3$, set $m = (p-2)/3$ and $L = m + 1 = (p+1)/3$. Retain the definitions of a_i, the strict prefix B_L and $U_p(L)$ from (46)–(49). Here the prefix is another direct application of Corollary 5: take $f(w) = 1 - w^3/2$ and $\lambda = 1$. The coefficients of f^h at w^{3i} are a_i, and $3m < p \leq 3L$. Thus $S_f(1) = B_L$. The coefficient H vanishes because $3 \nmid p - 1$; the exact trace is zero by the trace congruence and Hasse's bound. Since $f(1) = 1/2 \neq 0$, the argument is admissible, and the differentiated Miller query evaluates B_L in $O(\log p)$ field operations, without a point-counting search in this family.

The boundary a_L needed in (49) has a different arithmetic interpretation. Put $G = m! \bmod p$. Wilson reflection and the ratio of consecutive terms give

$$a_L = \frac{3}{2G^3}, \qquad G = \left(\frac{3}{2a_L}\right)^{(2p-1)/3} \quad \text{in } \mathbb{F}_p. \tag{52}$$

Indeed, $(2m)!L! = -1$ and $L! = LG$ give $(2m)! = -3G^{-1}$. Also $8^m = 1/2$, so $a_m = -6G^{-3}$, while $a_L/a_m = -1/4$. Cubing is bijective on $\mathbb{F}_p^\times$, yielding the inverse formula. Thus either a boundary routine or a modular-factorial routine supplies the other with $O(\log p)$ additional field operations.

The boundary in (52) also supplies the adjacent cutoff $m = L - 1$. The consecutive-term ratio and (49) give

$$a_m = -4a_L, \qquad B_m = B_L + 4a_L, \tag{53}$$

$$U_p(L) = 4B_L - \frac{17}{3}G^{-3}, \qquad U_p(m) = 4B_L - \frac{16}{3}G^{-3}. \tag{54}$$

In particular, $U_p(L) - U_p(m) = -G^{-3}/3 \neq 0$, so the adjacent sums cannot both vanish. Here $G \neq 0$ because $m < p$. The boundary multiplier $2L(2L+5) = 34/9$ vanishes only at $p = 17$ in this domain. Although the weighted sum then simplifies, the implementation still acquires and checks a_L and G separately.

The fast prefix therefore leaves a separate boundary-acquisition problem. Proposition 11 gives its full de Rham interpretation in an explicitly specified basis. Neither zero trace nor that interpretation supplies a fast boundary extractor. The computational companion describes exact boundary methods and the different gains available for a single prime and for a dense table across primes (Jordan, 2026e).

8 Scope and computational evidence

An active branch pole merges the two simple poles of the residue differential into a second-kind pole. Proposition 3 remains valid, but the nonbranch theorem does not provide the extra local coefficient required for a logarithmic branch query. The trace-free recurrence supplies an $\widetilde{O}(\sqrt{p})$ alternative. Higher-order poles, unsupplied denominator factorizations and arbitrary polynomial parts also require additional work. The included domain and the behaviour at infinity must always be specified.

The preferred coefficient evaluator in Corollary 5 permits $f(0) = 0$ at a nonzero nonbranch argument. This computational extension uses the pole as normalization origin; it does not identify the original residue construction at $(0, 1)$ when that point does not exist. Theorem 1 retains its own hypotheses.

Proposition 9 already holds over perfect extension fields. For $q_k = p^k$ and $\lambda \in \mathbb{F}_{q_k}^{\times}$, put $e_k = d^{(q_k-1)/2} \in \{1, -1\}$ and define $\tau_0 = 2$, $\tau_1 = \tau$, $\tau_k = \tau\tau_{k-1} - p\tau_{k-2}$. The divisor class is annihilated by $M = q_k + 1 - e_k\tau_k$. If $p \nmid M$, (31) gives $S_f(\lambda) = H + (r/M)(H\Psi_M(\mathfrak{D}) - \Psi_M(\mathfrak{D})^p)$, with a Miller chain of length $O(k \log p)$. Since $\tau_k \equiv \tau^k \pmod{p}$, the condition $p \mid M$ is equivalent to $e_k\tau^k \equiv 1 \pmod{p}$. The base-field simplification and equality with $T_f(1/\lambda)$ require separate arguments over extensions.

8.1 Higher genus

The residue cancellation behind Proposition 9 extends formally to a smooth curve of genus g: the holomorphic remainder after cancellation of the simple poles is written in a holomorphic basis as $\eta = \sum_{i=1}^{g} c_i\omega_i$, and $\mathcal{C}\eta = \sum_i c_i^{1/p}\mathcal{C}\omega_i$ involves the Cartier–Manin matrix. For a principal multiple of a divisor, logarithmic differentials and Jacobian transfer maps remain available (Rück, 1999). For $g > 1$, evaluation at the distinguished point supplies only one linear functional on this vector correction. Replacing H by a matrix alone therefore does not give the coefficient-sum formula or its exceptional normalization; these require further choices and arguments. No higher-genus evaluator is asserted here.

8.2 Correctness checks

The retained structural evaluator agreed with direct polynomial powers on 20,316 admissible queries over primes from 11 to 499, on 906 sampled models, including 368 exceptional queries and 2,512 split-algebra cases. A separate exhaustive check of 343 trace-one short models, for primes from 7 to 47, checked the high block against Miller constants at 11,730 nonidentity rational points. The characteristic-five tests in the earlier implementation were outside the stated prime range of Theorem 1; the uniform evaluator has a separate characteristic-five proof.

For the uniform-order implementation, 14,037 development comparisons and 5,400 confirmation comparisons agreed with direct character enumeration and the preceding corrected evaluator. Development included every squarefree cubic and quartic over $\mathbb{F}_5$, with nonzero leading coefficient and all nonbranch poles. An additional 47,712 nonbranch rational points checked Theorem 2 against polynomial powering. Confirmation used nine primes disjoint from that verifier's development prime set (Jordan, 2026d).

The supplied weighted adapter separately passed 3,840 coefficient-query comparisons, each checked for character, affine-point and projective-point output, for 11,520 output comparisons. The cases include zero-residue exclusions, constants and empty affine domains. The new truncation formula (21) was additionally checked by a separately implemented base-field chain against direct polynomial powers on 6,489 nonzero nonbranch arguments on 240 models over 15 primes from 5 to 59. These counts concern distinct checking procedures; overlapping edge-case counters are not additional independent samples. The source files, deterministic inputs and exact outputs accompany the revision. Finite agreement supplements the proofs.

For the present editorial revision, the independent coefficient verifier reproduced those 6,489 comparisons and the de Rham coefficient relations at 12 primes. A separate executable base-field adapter passed 6,693 single-pole and truncation queries on 240 development models. After freezing that adapter, a sample on nine disjoint primes passed another 11,905 of each query on 144 models. Direct enumeration of actual affine points also checked 6,912 weighted character, affine and projective outputs across the two samples. Branch queries were explicitly routed outside the fast adapter, and deliberately invalid inputs were rejected. The companion records exact counts, overlapping edge cases, candidate hash, environment and commands; these new checks do not revalidate every historical implementation or timing claim.

8.3 Measured workloads

The historical quarter-point pilot included specialized CM preparation in both complete outputs. Its seven inputs gave Cartier–Miller Python medians from 0.05 to 0.29 ms and Harvey/BGS C++ medians from 0.01 to 16.19 ms as the prime increased from 97 to 100,000,037. These measurements illustrate named implementations; the query bound alone does not determine the crossover, and the sequential nonresidue scan in the specialized preparation is not Schoof's worst-case algorithm. The complete original table, software versions and timing protocol remain in the computational companion.

The more general cold character-sum experiment charged exact-trace preparation to the historical Cartier–Miller implementation. Its medians are retained in Table 2; these timings have not been assigned to the newer uniform-order evaluator.

Table 2. Cold single-pole character-sum medians in milliseconds.

Within each cell, cubic / quartic. A dash means the method was not run at that class, not an extrapolated runtime. The native column is the best measured linear method in that class (C recurrence, C enumeration or FLINT polynomial power).

Bits	Historical Cartier–Miller including trace	Harvey/BGS adapter	Best native linear
10	0.138 / 0.169	0.561 / 0.610	0.033 / 0.040
20	0.246 / 0.239	6.34 / 14.78	25.04 / 38.93
30	0.367 / 0.356	170.21 / 474.16	—
40	1.18 / 0.983	6,900 / 16,000	—
62	131.73 / 133.69	—	—

The host reported an Intel Xeon at 2.10 GHz, two virtual CPUs and 7.8 GiB RAM. The software included CPython 3.13.16, PARI 2.17.2 without SEA data and the Harvey hypellfrob kernel. Methods ran sequentially, with validation outside the timed region. The recurrence comparator had a declared 40-bit ceiling. At 62 bits the separately sampled trace-only medians were about 122/117 ms for cubic/quartic cases; those are not an exact decomposition of the displayed cold medians.

Reuse changes the comparison. On the tested small curves, dense all-pole polynomial or correlation methods outperformed scalar loops, whereas some sparse workloads favoured the historical Miller evaluator. Fixed-domain coefficient-only changes reduced every method to the same cached dot product. A direct Voloch–Semaev adapter was faster than the historical candidate on known-trace exceptional inputs, with the same asymptotic short-chain cost. The computational companion retains these results, budget decisions, raw-data provenance and limitations. No new timing run of the preferred evaluator is claimed.

8.4 Limitations and further questions

The benchmark hosts were uncontrolled and the tested implementations, languages and trace providers differed. The measurements establish no universal crossover or superiority over all tools. Finite checks and AI feedback are not specialist proof review. The work is modulo p; a congruence modulo p^2 needs additional arguments. Projective arithmetic, exact trace frontends, branch coefficients, extension-field exceptional cases and higher-genus analogues merit separate treatment. No general computational lower bound for boundary acquisition is asserted.

AI assistance

Generative AI was used extensively in mathematical exploration, programming, computational checking and manuscript preparation under the author's direction. The reported checks arose within

that research programme and are not human peer review. The manuscript remains unrefereed by a human specialist.

A Cubic and quartic model changes

Use the quartic-to-Weierstrass construction with the rational point as origin (Connell, 1999, Section 1.2); the following formulas specify the version and scale required here. Write $f = 1 + f_1 w + f_2 w^2 + f_3 w^3 + f_4 w^4$, with $f_4 = 0$ for a cubic, and set

$$x = \frac{2\,(v+1) + f_1 w}{w^2}, \qquad y = \frac{4\,(v+1) + 2\,(f_1 w + f_2 w^2) - f_1^2 w^2/2}{w^3}. \tag{55}$$

Substitution gives the generalized Weierstrass equation with

$$a_1 = f_1, \quad a_2 = f_2 - f_1^2/4, \quad a_3 = 2f_3, \quad a_4 = -4f_4, \quad a_6 = a_2 a_4. \tag{56}$$

Near P_+, direct expansion gives

$$\begin{aligned} x &= 4w^{-2} + 2f_1 w^{-1} + f_2 - f_1^2/4 + O(w), \\ y &= 8w^{-3} + 4f_1 w^{-2} + \left(4f_2 - f_1^2\right) w^{-1} + O(1). \end{aligned} \tag{57}$$

Thus P_+ maps to the identity at infinity. The original points at infinity have affine images on this short model. The removable image of $(0, -1)$, used in the preferred evaluator, is specified in (18).

Put $b_2 = a_1^2 + 4a_2$, $b_4 = 2a_4 + a_1 a_3$, $b_6 = a_3^2 + 4a_6$, and use $X = x + b_2/12$, $Y = y + (a_1 x + a_3)/2$. The short equation is $Y^2 = X^3 + AX + B$, where

$$A = b_4/2 - b_2^2/48, \qquad B = b_6/4 - b_2 b_4/24 + b_2^3/864. \tag{58}$$

All denominators are units for $p \geq 5$. The invariant differentials are proportional; comparing their leading terms in (57) fixes

$$\omega = \kappa_\omega \frac{dX}{2Y}, \qquad \kappa_\omega = -2. \tag{59}$$

B The rounded third de Rham boundary

Throughout this appendix assume $p \geq 11$, $p \equiv 2 \pmod 3$, and put $h = (p-1)/2$, $L = (p+1)/3$ and $a_L = \binom{2L}{L} 8^{-L}$.

The rounded-third model $v^2 = 1 - w^3/2$ becomes $E : y^2 = x^3 + 1/4$ under $x = -w/2$, $y = v/2$. For the following cohomological statement use the point at infinity O as origin and fix $\omega = dx/y$; this differential has the opposite sign to dw/v under that coordinate change. Write $F = x^3 + 1/4$ and $F^h = \mathcal{T} + x^p \mathcal{S}$, with $\deg \mathcal{T} < p$. Let U be E with O removed, and W be E with $(0, 1/2)$ and $(0, -1/2)$ removed. A Čech–de Rham cocycle is (α_U, α_W, g) with $\alpha_U - \alpha_W = dg$, and a coboundary is $(du_U, du_W, u_U - u_W)$. Fix

$$e_1 = [(\omega, \omega, 0)], \qquad e_2 = \left[\left(x\omega, \frac{\omega}{2x^2}, \frac{2y}{x}\right)\right]. \tag{60}$$

Proposition 11 (Rounded third boundary in a specified basis). The classes (e_1, e_2) form a basis of $H^1_{\mathrm{dR}}(E/\mathbb{F}_p)$. With the Cartier convention of (6),

$$d(y\mathcal{S}) = a_L \omega, \qquad \mathcal{C}(\omega) = 0, \qquad \mathcal{C}(x\omega) = \mathcal{C}\left(\frac{\omega}{2x^2}\right) = 2a_L \omega. \tag{61}$$

Define $V_{\mathcal{C}}\left[(\alpha_U, \alpha_W, g)\right] = [(\mathcal{C}\alpha_U, \mathcal{C}\alpha_W, 0)]$ and $F_{\mathrm{abs}}\left[(\alpha_U, \alpha_W, g)\right] = [(0, 0, g^p)]$. These maps are linear over $\mathbb{F}_p$. With columns giving images of basis vectors,

$$[V_{\mathcal{C}}] = \begin{pmatrix} 0 & 2a_L \\ 0 & 0 \end{pmatrix}, \qquad [F_{\mathrm{abs}}] = \begin{pmatrix} 0 & -2a_L \\ 0 & 0 \end{pmatrix}. \tag{62}$$

Proof. For a polynomial P, put $\mathscr{D}(P) = FP\prime + F\prime P/2$, so $d(yP) = \mathscr{D}(P)\omega$. Since $h + 1/2 = 0$ in the field, $\mathscr{D}(\mathcal{T}) = -x^p\mathscr{D}(\mathcal{S})$. The expansion of F^h has exponents divisible by three, with $\deg \mathcal{T} = p - 2$, so $\deg \mathscr{D}(\mathcal{T}) \leq p$. Consequently $\mathscr{D}(\mathcal{S})$ is constant, equal to $\left[x^{p+1}\right] F^h/4 = a_L$. Cartier coefficient selection gives $\left[x^{p-1}\right] F^h = 0$, $\left[x^{p-2}\right] F^h = 2a_L$ and $\left[x^{p+1}\right] F^h = 4a_L$, proving (61).

The identity $d(2y/x) = (x - 1/(2x^2))\omega$ verifies the cocycle condition for e_2. Its first differential is regular on U: ω is regular everywhere and x has a pole only at O. The second differential is regular on W; at O it has order four because $\mathrm{ord}_O(x) = -2$ and ω has order zero. Both components and the transition function are regular on the required opens and overlap.

The class of $2y/x$ in $H^1(E, \mathcal{O}_E)$ is nonzero. Otherwise write $2y/x = u_U - u_W$ with u_U, u_W regular on the respective opens. Since u_W is regular at O, the polar part of u_U there would be the nonzero simple pole of $2y/x$. But u_U, a function regular on U, would then be a nonconstant element of $L(O)$, contrary to genus-one Riemann–Roch. For a smooth proper curve, the Hodge–de Rham exact sequence is

$$0 \longrightarrow H^0(E, \Omega^1_E) \longrightarrow H^1_{\mathrm{dR}}(E) \longrightarrow H^1(E, \mathcal{O}_E) \longrightarrow 0. \tag{63}$$

The first nonzero class is represented by e_1, and the image of e_2 is the nonzero class just proved. As each outer space has dimension one, e_1, e_2 form a basis. This uses the standard proper-curve de Rham framework (Oda, 1969).

Cartier annihilates exact differentials, so the local Cartier images in a cocycle agree on the overlap; this also shows that $V_{\mathcal{C}}$ is well defined on cohomology. Raising the transition function to the pth power annihilates its differential and carries coboundaries to coboundaries, so F_{abs} is well defined. Both kill e_1. Formula (61) proves the displayed image of e_2 under $V_{\mathcal{C}}$.

For F_{abs}, write $(2y/x)^p = u_U - u_W$, where $u_U = 2y\mathcal{S}$ and $u_W = -2y\mathcal{T}/x^p$. The first is polynomial in affine coordinates and hence regular on U. The second is regular on the finite part of W, because $x = 0$ is excluded. At O, using $\deg \mathcal{T} = p - 2$, its order is

$$\mathrm{ord}_O(y\mathcal{T}/x^p) = -3 - 2(p - 2) + 2p = 1, \tag{64}$$

so it is regular there too. Since $du_U = du_W = 2a_L\omega$, subtracting this coboundary from $(0, 0, (2y/x)^p)$ leaves $-2a_L e_1$. This proves (62) without an auxiliary companion proof.

The usual genus-one Cartier–Manin matrix on regular differentials is the single entry zero. The nonzero entry $2a_L = 3G^{-3}$ in (62) is a matrix entry of the full de Rham operators in this specified basis. If e_1 and e_2 are replaced by ae_1 and be_2, with $a, b \in \mathbb{F}_p^\times$, this entry becomes $(b/a)2a_L$. It is not asserted to be an intrinsic scalar independent of the chosen differential and basis. Although $a_L\omega$ is exact in the function field, it is nonzero in projective de Rham cohomology: its rational primitive has poles and cannot be made globally regular. This is a global regularity obstruction, not a prohibition on cancelling a pole at one point. Neither the zero trace nor this interpretation provides a fast boundary extractor, and no period normalization is needed for the statement.

C The two pole implementation

The structural scalar can also be computed directly on the original model. This alternative explains the earlier implementation and its benchmark provenance; Corollary 5 is the recommended coefficient evaluator. Since $[M_e]\, R_+ = [M_e]\, R_-$, the quotient $f_{M_e,R_+}/f_{M_e,R_-}$ has divisor $M_e\mathfrak{D}$. Its pole terms at O cancel, and $dX/(2Y) = \left(1 + O\left(z^4\right)\right) dz$, so

$$\Psi_{M_e}(\mathfrak{D}) = \frac{\alpha_{M_e}(R_+) - \alpha_{M_e}(R_-)}{\kappa_\omega}. \tag{65}$$

Compute without finding a square root in the two-dimensional étale algebra $\mathcal{A}_d = \mathbb{F}_p[\xi]/\left(\xi^2 - d\right)$. Run the chain with $v_0 = \xi$ and write its constant as $\alpha = A_0 + B_0\xi$. Algebra conjugation gives the other chain. Equations (4) and (65) become

$$r\Psi_{M_e}(\mathfrak{D}) = \frac{2B_0}{\kappa_\omega \lambda} = -\frac{B_0}{\lambda}, \qquad S_f(\lambda) = H + e\frac{B_0}{\lambda}. \tag{66}$$

For $e = -1$, the algebra is a field; for $e = 1$, it is a product of two fields. The latter requires branch tests componentwise. An element $u = A_0 + B_0\xi$ with norm zero and $u \neq 0$ has $B_0 \neq 0$ and supplies the root $(A_0/B_0)^2 = d$. Split the saved state by $\xi \mapsto \pm A_0/B_0$ and resume the remaining binary steps in both fields. At addition, first handle an identity point; for equal abscissae use a vertical step when ordinates are opposite, and a tangent when the points coincide, treating an ordinate of zero as vertical. Otherwise use the chord. At doubling, an identity or ordinate-zero point is a zero-contribution step; otherwise use the tangent. A nonzero zero divisor in an equality test can also split the state before choosing these branches, so a tangent in one component and a vertical addition in the other are handled separately. Splitting can occur only once. This is the degree-two instance of dynamic evaluation, or the D5 principle (Della Dora et al., 1985).

Run Algorithm 1 over this quadratic algebra, splitting an uncommitted operation when an equality test or inverse supplies a nonzero zero divisor. Specialize the saved point and accumulator states into the two factors and resume the same operation in each field, applying its accumulator update exactly once. Identity and vertical steps use the same conventions as the base-field chain. At most one split occurs. The alternative retains an $O(\log p)$ field-operation bound, but neither it nor the exceptional-normalization proof is needed to execute the preferred twist-point query.

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